Scalable AI Oversight with Multiple Verifiers

Can a panel of small judges outwit one big one?

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The Core Problem

Why oversight is hard

As AI systems become more capable, they may produce answers that **look correct** to a human (or automated checker) but are actually wrong.

- A **prover** (the AI) generates answers to math problems.
- A **verifier** (the judge) scores how trustworthy each answer looks.
- The prover is rewarded for *fooling* the verifier — getting a high trustworthiness score even when the answer is wrong.

The question

How do we build a verifier that is hard to fool, even by a prover specifically trained to fool it?

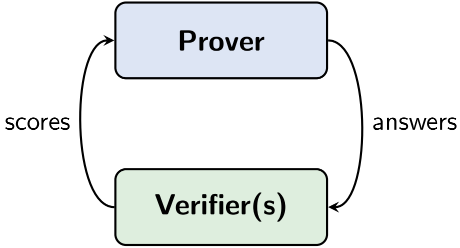
The Setup: A Game Between Prover and Verifier

Two roles, two goals:

- **Prover** — a language model fine-tuned to solve math. In "sneaky" mode it tries to produce wrong answers that the verifier scores as correct.
- **Verifier** — a language model fine-tuned to assign a *trustworthiness score* $f \in [0, 1]$ to each answer.

They train against each other in rounds:

1. Verifier trains to distinguish correct from incorrect.
2. Prover trains to fool the updated verifier.
3. Repeat.



Our Method: A Panel of Small Verifiers

Baseline (Kirchner et al. 2024)

One large verifier (7B parameters) judges each answer. If the prover learns its blind spots, it can reliably fool it.

Our approach: Multi-Verifier with PE-Min

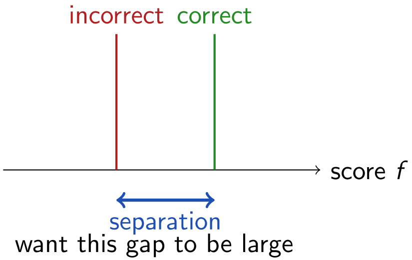
- Use **3 smaller verifiers** (3B parameters each), each specialised on a different aspect: *logical structure*, *computational accuracy*, *problem alignment*.
- The **oversight score** is the **minimum** across all three verifiers ("PE-Min" rule) — the answer must convince *every* judge.
- Verifiers are trained with a **pairwise ranking loss** (Bradley-Terry) so they learn *which* answer is more trustworthy, not just a raw score.

Intuition: Exploiting one judge's blind spot doesn't help if the other two catch you.

How We Measure Success

Key metrics:

- **AUROC** — How well does the verifier rank correct answers above incorrect ones? (1.0 = perfect, 0.5 = random)
- **Separation** — Average score gap between correct and incorrect answers. Bigger = more calibrated.
- **Binary accuracy** — What fraction of answers does the verifier classify correctly?
- **Kirchner attack** — We fine-tune a fresh prover specifically to fool the trained verifier. What fraction of wrong answers does it get accepted? Lower = more robust.



Results

Metric	Multi-3B (ours)	Single-7B (baseline)
AUROC	**0.827**	0.779
Score separation	0.176	**0.312**
Binary accuracy	**73.8%**	72.5%
Helpful correctness	**42.5%**	40.0%
CGC attack — % wrong accepted	**81.7%**	90.0%
SRC attack — % wrong accepted	**63.3%**	68.3%

Table: Lower attack acceptance = harder to fool = better.

Take-away: Multi-verifier wins on detection quality (AUROC) and is harder to fool under adversarial attack, at the cost of less-calibrated absolute scores.

What's Next

- Current results use a **supervised** oversight rule. We are running the full **PE-Min** variant (minimum-across-verifiers) to see if it is even harder to fool.
- Extend to **more training rounds** so the prover accumulates adversarial strategies over time — the key dynamic from Kirchner et al.
- Produce a direct apples-to-apples comparison with the Kirchner paper numbers on the same dataset and attack budget.

Hypothesis

With PE-Min, the prover must fool *all three* verifiers simultaneously. This should drive the attack acceptance rate below the single-verifier baseline, giving us stronger scalable oversight at lower parameter cost.